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Social implementation and intervention with estimated morbidity of heat-related illnesses from weather data: A case study from Nagoya City, Japan

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ABSTRACT

The estimation of heat-related illness cases is a key factor in proposing and implementing suitable intervention strategies and healthcare resource management. This paper proposes new frameworks to estimate the number of patients with heat-related illnesses by administrative wards in Nagoya City using 2014–2019 data. The proposed frameworks are based on the derivation of estimation formulae and machine learning. The daily residual estimation error in the 16 wards was less than one person with both the frameworks. The daily working time average ambient temperature may yield a better correlation than the daily average temperature or daily highest temperature with the number of patients transported by an ambulance from outdoor sites. The results also indicate that patients transported from indoor sites are influenced by earlier ambient conditions over approximately 50 days. In contrast, those transported from outdoor sites are influenced by a relatively short period (20 days), which may correspond to heat adaptation. The frameworks provide a better understanding of the different factors that would lead to an accurate prediction of the number of cases of heat-related patients from weather forecasts. These findings would lead to efficient ambulance allocation as well as public awareness on hot days to suppress heat-related morbidity.

1. Introduction

The mortality and morbidity of heat-related illnesses have increased in many countries as a result of climate change and global warming. In developed countries, increases in mortality and morbidity have been reported with an increasing frequency and intensity of heat waves (Chen et al., 2017; Conti et al., 2005; Khare et al., 2015; Meehl & Tebaldi, 2004; Williams et al., 2018). In developing countries, mortality from heat-related diseases has also been reported (Nori-Sarma et al., 2019; Rauf et al., 2017). In addition, the global population is rapidly aging (He et al., 2016; WHO, 2020). The number of elderly individuals aged 60 and over has exceeded globally and will more than double by 2050 (WHO, 2020). Population ageing increases the mortality and morbidity of heat stroke (Allen & Segal-Gidan, 2007; Flynn et al., 2005). Therefore, the risk assessment of heat stroke is an essential issue that needs to be addressed for global sustainability.

The effect of extreme ambient conditions on the mortality and morbidity of heat-related illnesses is of significant concern in the field of public health. Many studies have indicated a significant correlation between morbidity/mortality and ambient conditions (Conti et al., 2005; Hua et al., 2021; Madrigano et al., 2015; Meehl & Tebaldi, 2004; Nakai et al., 1999) and between ambient temperatures and the number of emergency ambulance dispatches (Bassil et al., 2011; Kotani et al., 2018; Sangkharat et al., 2020). The primary index used for the number of patients with heat-related illnesses is the peak value of the daily ambient temperature or wet-bulb globe temperature (WBGT) (Budd,

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2008; Lemke & Kjellstrom, 2012; Parsons, 2006). WBGT, which considers temperature, humidity, and radiation heat (Parsons, 2006; Zhang et al., 2014), is a global index defined by the International Organization for Standardization (ISO) for heat stroke prevention (Bröde et al., 2012). This index is often represented as the heat load at a certain time (e.g., the daily maximum value). In contrast, human physiological states change over time, depending on the heat load (L. Li & Zha, 2020; Ministry of the Environment, 2018).

The physiological parameters linked to heat-related illnesses include core temperature elevation and water loss (Epstein & Yanovich, 2019), which are related to dehydration and heat balance collapse, respectively. In this context, the American Conference of Government Industrial Hygienists (ACGIH) standard recommends preventing body core temperature elevation above 1°C. It also provides guidelines for physical activity and water intake (ACGIH, 2017). Chen *et al.* (Chen et al., 2017) reported that prolonged heat exposure could have added adverse health impacts beyond risk owing to higher daily temperatures. However, most studies have discussed the correlation between instantaneous values, whereas the lag effect of ambient temperature on the morbidity of heat-related illnesses has been reported in only a few recent studies (Cui et al., 2020; Kotani et al., 2018).

In Japan, the number of transported people and patients closely match (Shimazaki et al., 2020) because ambulance transportation is free. Ambulances are managed in the administrative area; thus, accurate data on morbidity is available, unlike in other developed countries. According to the data, records of the number of people transported by ambulance owing to heat-related illnesses have been kept since 2008. The criteria/definition for heat-related illness was revised in 2013 (Fire and Disaster Management Agency). The number of transported patients reached 95,137 in 2018 and has since remained high (Fire and Disaster Management Agency). Many patients experience symptoms of illness while at home. In particular, more than 50% of the patients with heat-related illnesses are the elderly who suffer at home, whereas only 30% or less of the young adult patients suffer at home (Fire and Disaster Management Agency, n.d.). Many studies have reported that elderly citizens are the most vulnerable to the consequences of potential heat waves (Khare et al., 2015; Kravchenko et al., 2013; M. Li et al., 2015; Oudin Åström et al., 2011) because of the decline in thermal sensitivity and sweating rate capacity (Dufour & Candas, 2007; Hirata et al., 2015).

Heat-related illnesses have been classified into the following two main categories: classical/non-exertional heat stroke and exertional heat stroke. The majority of patients in the former are elderly, and most scenarios are largely similar (Coris et al., 2004; Yeo, 2004). The latter is primarily characterized by environmental factors and extensive workloads and occurs primarily outdoors. The potential cumulative effect of heat exhaustion, that is, the effect of weather conditions over several days, is suggested to influence the number of patients of heat-related illnesses (Cui et al., 2020; Kotani et al., 2018; Wang et al., 2019). This effect may be attributed to heat adaptation (Bakhsh et al., 2018; Hori, 1995; Lee & Sheridan, 2018). The negative health impacts of heat can be predicted and largely prevented by specific public health actions, and it is especially important to recognize which individuals are at high risk, use appropriate hydration, and consider acclimatization and environmental conditions (Glazer, 2005; Hifumi et al., 2018). Consequently, communication and intervention strategies to reduce heat-related mortality and morbidity have become important issues for national/regional administration (Glazer, 2005; Kravchenko et al., 2013).

Models have been proposed that predict the risk or impact of heat-related illnesses using broadcasting via social networks (Fuse et al., 2019; Hondula et al., 2021) or gross domestic product values (Wang et al., 2019). However, these data could not be easily obtained, and thus, it is difficult to predict the number of patients in the near future. In our previous study, we formulated for the first time an equation to estimate the number of elderly patients with heat-related illnesses in terms of weather data and population distribution, in addition to the effect of heat adaptation (Kodera et al., 2019). The features of our equations are the necessity of only a few publicly available input values. The limitation of the equation was that the number of patients with exertional heat stroke (especially for younger adults) was not considered. In addition, the end of the rainy season is used as a factor for expressing short-term heat adoption, which is a unique characteristic of Japan. If a system can be developed to estimate the total number of heat-related illness patients using fewer meteorological factors, it will be useful for emergency medical services and improve the efficiency of ambulances, health services, and the environment (Macintyre & Heaviside, 2019; Taylor et al., 2018).

Fig. 1. Aichi prefecture and Nagoya city on the map of Japan.

This study proposes a more general framework for the daily estimation of the number of patients with heat-related illnesses. We collected data in Nagoya City, Japan, for validation and assessment of the proposed framework. We formulate a new equation that considers the differences in the mechanisms of heat stroke, that is, non-exertional and exertional heat stroke. Moreover, we consider the design of a machine learning architecture for validation and verification. To the best of the authors' knowledge, this is the first study to provide frameworks to estimate the morbidity of heat-related illnesses in administrative districts for practical application in ambulance allocation. These frameworks also enable specific alerts for each residential area to be linked to weather forecasts for public awareness.

2. Methods

2.1. Data sources

The region considered here is Nagoya City, one of the primary cities of Aichi prefecture in Japan, located at 35°10' N and 136°54' E (see Fig. 1). The city's population is approximately 2.3 million, and it is the third largest city in terms of population, excluding 23 wards of Tokyo Metropolis in Japan (Ministry of Internal Affairs and Communication). Three datasets were used in this study. The first consisted of records of emergency transport by ambulance pertaining to heat-related illness obtained by the authors (Nagoya City Fire Department), which approximates the number of heat-related-illness patients. The dataset was used as an objective variable. The dataset included the date of incidence, location, and age of patients over 18 years old, who accounted for 90.8% of all cases. The distinction between exertional and non-exertional heat stroke was intuitively determined based on the location and other related recorded data of the incidence.

The second dataset consists of weather data for Nagoya City. The dataset was used as an input variable because it is closely associated with heat-related mortality and morbidity (Kodera et al., 2019; Kotani et al., 2018; Sangkharat et al., 2020). The ambient temperature was provided by the Japan Meteorological Agency (JMA) and the hourly WBGT by the Ministry of the Environment in Japan (Ministry of the Environment). The third dataset comprised the age composition of the

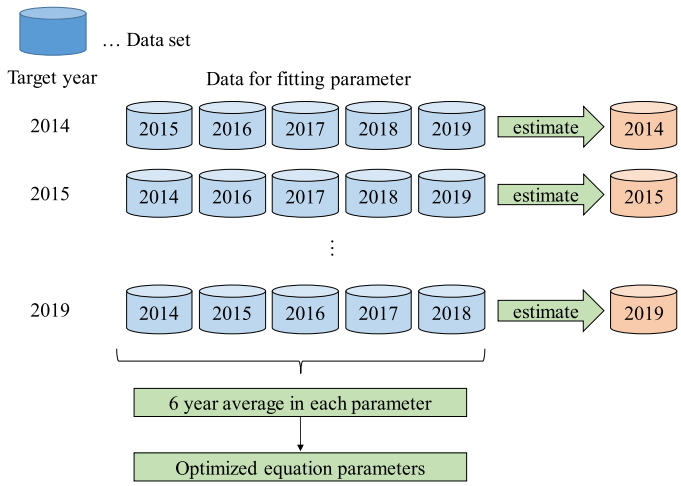

Fig. 2. Parameter fitting process used in this study.

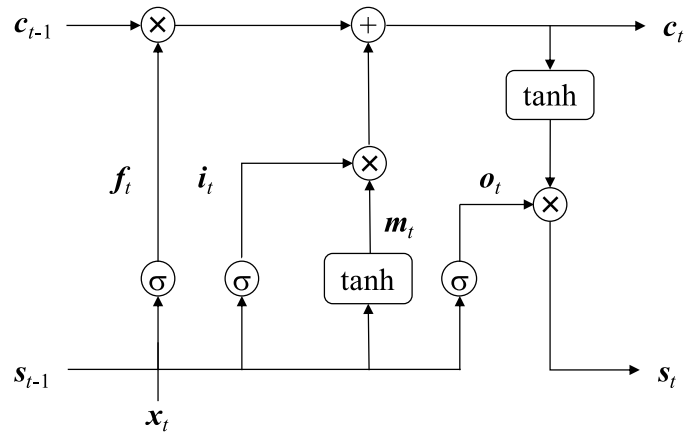

Fig. 3. Architecture of single LSTM node.

population in each administrative ward and the entirety of Nagoya City, provided by the Statistics Bureau of Japan (Nagoya City, 2015). It was used for considering the difference in risk of heat-related illness with age. The definition of "ward" is based on the administrative definition of administrative regions located in ordinance-designated cities of Japan (Ministry of Internal Affairs and Communication).

2.2. Equation formulation

In our previous study (Kodera et al., 2019), we presented an equation to estimate the number of elderly patients with heat-related diseases under ambient temperature for three consecutive days. Here, we extend the equation to estimate non-exertional/exertional heat stroke patients separately. Two equations were developed based on the locations where the individuals made emergency calls (indoor or outdoor). In addition, heat adaptation was assumed to be the weighted average value of the ambient parameters.

The proposed equations are as follows:

$$y(x) = y_{indoor}(x) + y_{outdoor}(x), \tag{1}$$

$$y_{indoor}(x) = a_{in}\cdot\left[e^{k_{in}\cdot(0.6x_0+0.2x_1+0.2x_2)} + l_{in}\right]\cdot\sum_n\{P(n)\cdot(be^{cn}+d)\}, \tag{2}$$

$$y_{outdoor}(x) = a_{out}\cdot\left[e^{k_{out}\cdot x_0} + l_{out}\right]\cdot\sum_n\{P(n)\cdot(be^{cn}+d)\}, \tag{3}$$

$$k_{in,out} = f_{in,out}\cdot\left(\sum_{r=1}^{R} w_r\cdot T_r\right) + g_{in,out}, \tag{4}$$

where y_{indoor} and $y_{outdoor}$ denote the daily number of patients with heat-related illness transported from indoor and outdoor locations, respectively. x_i is a metric related to ambient conditions (ambient temperature or WBGT) within the past i consecutive days. In Eq. (2), the heat-related-illness patients transported from indoors are affected by the daily average temperature for the last three consecutive days; the weightings of x_0, x_1, and x_2 are 60%, 20%, and 20%, respectively (Kodera et al., 2019). In Eq. (3), the patients transported from outdoors are assumed to be affected by the weather on the same day (Ministry of the Environment, 2010), and the best input variables for outdoor estimation were investigated (see Section 3.1 for optimized input).

Parameter n denotes the age category (five-year age intervals from age 20 onward; $n = 1$ (20–24 years old), …, 14 (85 years old and over)); $P(n)$ denotes the population distribution by age in the entire or individual administrative district ward of Nagoya City (Nagoya City). Parameters b (= 0.171), c (= 0.494), and d (= 190.7) represent constants determined by the least-squares fitting method, as shown in Fig. 4 in Kodera et al., 2019, which expresses the increased risk with age. To estimate the number of patients in each ward, the estimate for the entire city of Nagoya was distributed according to each ward's weight in the population distribution.

In Eq. (4), to simulate the short-term heat adaption, the risk of heat-related illnesses k (coefficient of the input variable x_i) was assumed to decrease linearly with an increasing weighted daily average temperature over the past few tens of days (Pandolf, 1998) (see Section 3.1 below for parameter optimization). T_r denotes the daily average temperature r days before the target date, and w denotes the weighted linear function; f (slope) and g (intercept) characterize the ratio of linear reduction in the number of transported individuals according to the weighted daily average temperatures over the previous r days. Parameters a, k, l, f, and g were determined for patients transported from indoor and outdoor locations.

First, a and l were determined; k, f, and g were then computed via least-squares fitting. Finally, the value of each parameter was averaged over a six-year time frame (see Fig. 2). After setting parameters a, l, f, and g, this parameter fitting procedure was iteratively conducted for convergence. For example, f and g calculated in the previous step were fixed, and a and l were calculated. Similarly, a and l in the previous step were fixed, and f and g were recalculated.

Five years of data (extracted from 2014 to 2019) were used to determine the parameters in Eq. (1). Subsequently, data for the remaining year were used through a leave-one-out cross-validation study (Berrar, 2019); for example, data from 2014 to 2018 were used to determine the parameters for estimating the number of people transported in 2019. The estimated and actual number of individuals transported were compared to determine the effectiveness of the proposed method. Two metrics, ambient temperature and WBGT, were considered.

2.3. Recurrent neural network and random forest regression

A time-series-enabled neural network estimation is selected because the number of heat-related illness patients has a significant correlation with environmental factors, such as temperature and humidity, not only in the duration of the estimation date but also earlier (Kodera et al., 2019). The recurrent neural network (RNN), a deep learning approach that can consider the recurrence relationships between time-sampled data, is potentially useful (Schuster & Paliwal, 1997). However, the application of RNNs is challenging for long-term memory retention, which is limited owing to the gradient loss problem. A long short-term memory (LSTM) network solves this problem by incorporating memory units (Tian et al., 2018) and learns when to forget previous data and update the memory when necessary.

Table 1
LSTM network model structure.

	Network Layer	Number of Units
1	Input Layer	60
2	Hidden Layer I (LSTM)	60
3	Hidden Layer II (LSTM)	30
4	Fully Connected Layer	30
5	Output Layer	1

LSTM is a neural network structure that includes a memory gate and forget gate that can be trained to learn order dependency between items in a sequence. Unlike most other deep learning architectures, LSTM can learn the context of data (with a specified timeframe) more than the value of data (at an instance). Therefore, the LSTM was adopted in this study. The structure of the LSTM memory unit is illustrated in Fig. 3. The measured data are formulated into a multivariate time series with Q variables of length T as $\boldsymbol{X} = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_T)$, where $\boldsymbol{x}_t$ denotes the t-*th* data element of all the variables and x_t^q denotes the data of the q-th variable of $\boldsymbol{x}_t$. The main feature of the LSTM model is a memory cell $\boldsymbol{c}_t$ that considers information at time frame t on the time-series data obtained in this step. The cell is controlled by multiple gates and can either keep or reset the value depending on the state of the gate. Four gates are adopted to control whether to forget the current cell value (forget gate $\boldsymbol{f}_t$), to read its input (input gate $\boldsymbol{i}_t$), to output the new cell value (output gate $\boldsymbol{o}_t$), and to remember the current cell value (memory gate $\boldsymbol{m}_t$). The gates and cell are updated and the outputs are calculated as follows:

$$\begin{aligned}
\boldsymbol{i}_t &= \sigma(\mathbf{W}_{xi}\boldsymbol{x}_t + \mathbf{W}_{si}\boldsymbol{s}_{t-1} + \boldsymbol{b}_i),\\
\boldsymbol{o}_t &= \sigma(\mathbf{W}_{xo}\boldsymbol{x}_t + \mathbf{W}_{so}\boldsymbol{s}_{t-1} + \boldsymbol{b}_o),\\
\boldsymbol{f}_t &= \sigma(\mathbf{W}_{xf}\boldsymbol{x}_t + \mathbf{W}_{sf}\boldsymbol{s}_{t-1} + \boldsymbol{b}_f),\\
\boldsymbol{m}_t &= \tanh(\mathbf{W}_{xm}\boldsymbol{x}_t + \mathbf{W}_{sm}\boldsymbol{s}_{t-1} + \boldsymbol{b}_m),\\
\boldsymbol{c}_t &= \boldsymbol{f}_t \circ \boldsymbol{c}_{t-1} + \boldsymbol{i}_t \circ \boldsymbol{m}_{t-1},\\
\boldsymbol{s}_t &= \boldsymbol{o}_t \circ \tanh(\boldsymbol{c}_t),
\end{aligned} \tag{5}$$

where denotes the Hadamard product, the $\boldsymbol{W}$ matrices are the network weighting parameters, σ() is the sigmoid function, and $\boldsymbol{s}_t$ is the hidden state. Because these gates address exploding or vanishing gradients, LSTM networks are robust. The details of the LSTM network model structure used in this study are listed in Table 1.

The input variables are the ward label of Nagoya City m, daily average temperature, and working time (9:00–17:00) average temperature. These parameters are selected to correspond to the proposed equation. Initial study demonstrates that LSTM trained with 60 consecutive days provides the best network performance. The output variable was the daily number of heat-related-illness patients recorded in each ward. Training was conducted using the Adam algorithm as a network optimizer with learning rates = 0.001, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 0.00001$. The number of epochs and batch sizes were 100 and 2, respectively. We considered a leave-one-out cross-validation (Berrar, 2019), where a single year was excluded from the data, and the remaining were used for the training.

To validate the effectiveness of the LSTM model, we also conducted a comparison study using random forest (RF) regression—a common machine learning technique (Breiman, 2001). The same training dataset was used to train a set of decision trees optimized for the best performance.

3. Results

3.1. Parameter estimation

The metrics that had the highest correlation with the number of patients transported from outdoor locations were investigated. Because the time course of the physiological response is essential to characterize heat-related illnesses, the daily average, maximum, and working time average (9:00–17:00) temperatures were considered for the two metrics (ambient temperature and WBGT). The rationale for selecting this time slot (9:00–17:00) is that approximately 70% of the employed individuals in Japan work during this period (Broadcasting Culture Research Institute). The coefficients of determination for each metric are presented in Table 2. The coefficient of determination with the working time average temperature was the highest; however, it was comparable to that of the daily average. Therefore, in the following discussion, the average working time ambient temperature was used to estimate the number of patients transported from outdoor locations.

Then, the optimal number of days for simulating short-term heat

Table 2
Coefficient of determination over six years (2014–2019) with the daily average, highest, and working time (9:00–17:00) average value for each metric with proposed equation in outdoor estimation.

	Daily average	Daily highest	Working time average
Ambient temperature	0.671	0.647	0.680
WBGT	0.604	0.543	0.655

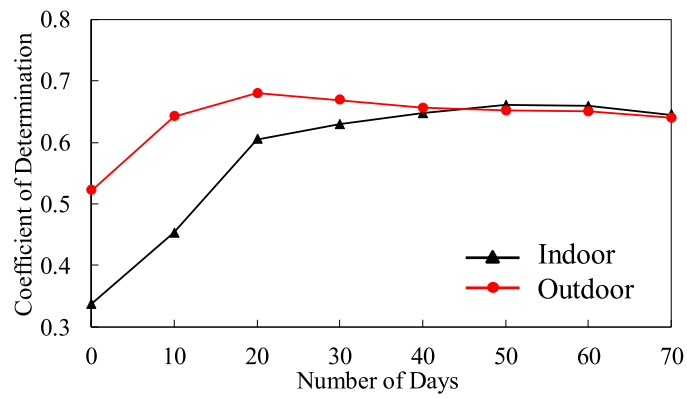

Fig. 4. Variation of coefficient of determination with number of days over which weighted daily average temperature is calculated. The horizontal axis corresponds to R in Eq. (4).

Table 3
Parameters of equations for estimating patients transported from indoor and outdoor locations.

	Indoor	Outdoor
a	1.23×10^{-5}	2.44×10^{-6}
l	-1.01×10^{3}	-2.03×10^{3}
f	-2.70×10^{-3}	-3.39×10^{-3}
g	0.33	0.37

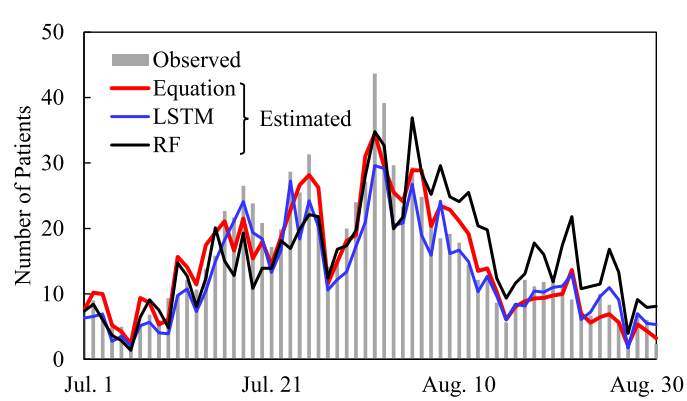

Fig. 5. Observed and estimated number of patients in Nagoya (averaged over 2014–2019).

Table 4
Coefficient of determination over six-year period using the proposed equation, LSTM, and RF.

	Proposed equation	LSTM	RF
2014	0.752	0.879	0.645
2015	0.866	0.896	0.810
2016	0.680	0.776	0.494
2017	0.586	0.610	0.340
2018	0.808	0.951	0.404
2019	0.849	0.685	0.587
Average	0.757	0.799	0.547

adaption was investigated for the estimation using the proposed equations (corresponding to R in Eq. (4)). Fig. 4 presents the dependence of the coefficient of determination on the number of days over which the weighted average ambient temperature was calculated, representing the heat adaptation. The determination coefficient increased with an increasing number of weighting days and then became constant. The peak coefficient of determination was observed for the indoor and outdoor patient estimations with 50 and 20 days of weighted days, respectively. The parameters used to estimate the number of patients transported from indoor and outdoor locations are listed in Table 3.

In general, these two factors affect one another, thereby requiring an iterative procedure. The difference in this procedure was marginal; thus, the original values were listed without further adjustment.

3.2. Validation study

Fig. 5 shows the observed and estimated number of heat-related illness patients averaged over six years (2014–2019) in the entire city of Nagoya. The proposed equations, RF, and LSTM can achieve a high matching estimation with the true values throughout the summer. As shown in Fig. 5, LSTM estimation is superior to RF, which is attributable to the nature of the time-series feature extraction of LSTM. Slight fluctuations can be observed only during the well-known Japanese "Obon" holiday (from August 8 to August 16). The daily residual error in the entire city of Nagoya calculated using the proposed equations, RF, and LSTM, were 5.16, 9.04, and 4.19 people, respectively. The coefficients of determination over six years for the proposed equation, RF, and LSTM are listed in Table 4. Both the residual error and the coefficient of determination indicate that the proposed equation and LSTM are better than RF for estimating the number of patients with heat-related illness.

3.3. Estimation at ward scale

Fig. 6 shows the estimates in each ward of Nagoya City compared to the actual values. The estimation of the number of patients using the equation was determined using the weighted average of the population. As shown in Fig. 6, the prediction by the LSTM was more consistent with the actual number than that using the proposed equation because in the proposed equation, the number of patients was estimated over the entire city of Nagoya and then distributed to each ward based on their population, whereas LSTM was trained using ward-wise data, which provides a more reliable estimation. Table 5 lists the daily residual errors over six years in each ward. The mean values of the 16 wards of the daily residual error over six years were 0.74 and 0.67 people for the proposed equation and LSTM, respectively. The highest error was observed in the Nakamura ward in both the proposed equation and LSTM, which is the central ward of Nagoya city and one of the most prosperous commercial areas.

4. Discussion

In this study, we developed the following frameworks to estimate the daily number of heat-related-illness patients in each ward by classifying patients according to indoor and outdoor locations: nonlinear regression equations and the LSTM model. To validate the LSTM, an RF model was constructed. Unlike in our previous study, in which only elderly patients were considered, the entire population was considered in this study. In addition, we assumed that the occurrence of heat-related illness indoors and outdoors could be approximated as non-exertional and exertional heat stroke, respectively, and we formulated two equations to consider their differences. Non-exertional heat stroke is common in the elderly (Japan Emergency Medicine Society, 2014), whereas exertional heat stroke is common in young adults or workers (Hua et al., 2021). The ambulance data of Nagoya City in 2014–2019 indicates that the elderly comprised 71% of patients transported from indoor locations, and young adults (20–64 years old) accounted for 55% of patients transported from outdoor locations.

First, we investigated the optimal input variable to estimate the number of patients transported from outdoor locations with the daily average, daily highest, and working time average values of two metrics (ambient temperature and WBGT) for the regression equations. As shown in Table 2, the optimal input variable is the average ambient temperature over the working time. To prevent heat-related illnesses, it is necessary to consider working hours. In addition, it was a useful result for practical implementation, where the same coefficient of determination was confirmed using the daily mean ambient temperature available online.

Fig. 4 indicates that the number of days to reach a steady state for the coefficient of determination in estimating the patients transported from indoor locations is 30 days longer than that for patients transported from outdoor locations. This result suggests that individuals transported from indoor locations are influenced by past ambient conditions over a long period (50 days), whereas those transported from outdoor locations are influenced by ambient conditions over a relatively short period (20 days) for short-term heat adaptation. The latter value corresponds to a study in which athletes required approximately two weeks or less for heat adaptation (Garrett et al., 2011). A linear weighted average was selected for the heat adaptation expression. Although not shown here, the difference in weighted average methods, such as exponential, arctan weightings, and simple averaging, marginally affected the coefficient of determination. For example, when selecting simple averaging, the coefficient of determination decreased to 0.750, which is nearly the same as that in this study.

Next, we constructed the RF and LSTM models and compared their estimation accuracy with that of the estimating equation. As shown in Fig. 5 and Table 4, the equation proposed here provides a high-accuracy estimate of daily heat-related illness occurrences in the entire city of Nagoya with an accuracy comparable to that of the LSTM model and higher than that of the RF model. Focusing on the estimation in each ward, the daily average residual error of 16 wards was also less than one person with both the proposed equation and LSTM (see Fig. 6 and Table 5). The prediction accuracy of LSTM exceeded that of the proposed equation in certain wards. Because the study was conducted in urban areas, considering the inflow and outflow of the daytime population will further improve the prediction accuracy of the estimation equation (see Appendix A).

The proposed equation is easy to use, making it ideal for simple daily alerts (public awareness) for heat-related illnesses to parents and educators (Rauf et al., 2017). The frameworks using LSTM is suitable for information, such as that used for emergency medical services (EMS), which requires more accurate predictions for ambulance allocation. These frameworks enabled specific alerts for each residential area linked to the weather forecast (see Appendix B). Various mitigation measures against heat stroke have been utilized globally (e.g., Fisk, 2015; Wong et al., 2021). They may serve as indicators for evaluating the applicability of these precautions. Furthermore, the proposed frameworks can be used to estimate the increase in the EMS requests per ward in advance. It will be possible to form appropriate EMS units and potentially save more lives by providing information regarding the areas where more EMS units should be dispatched. The frameworks can estimate the future number of patients with a heat-related illness by

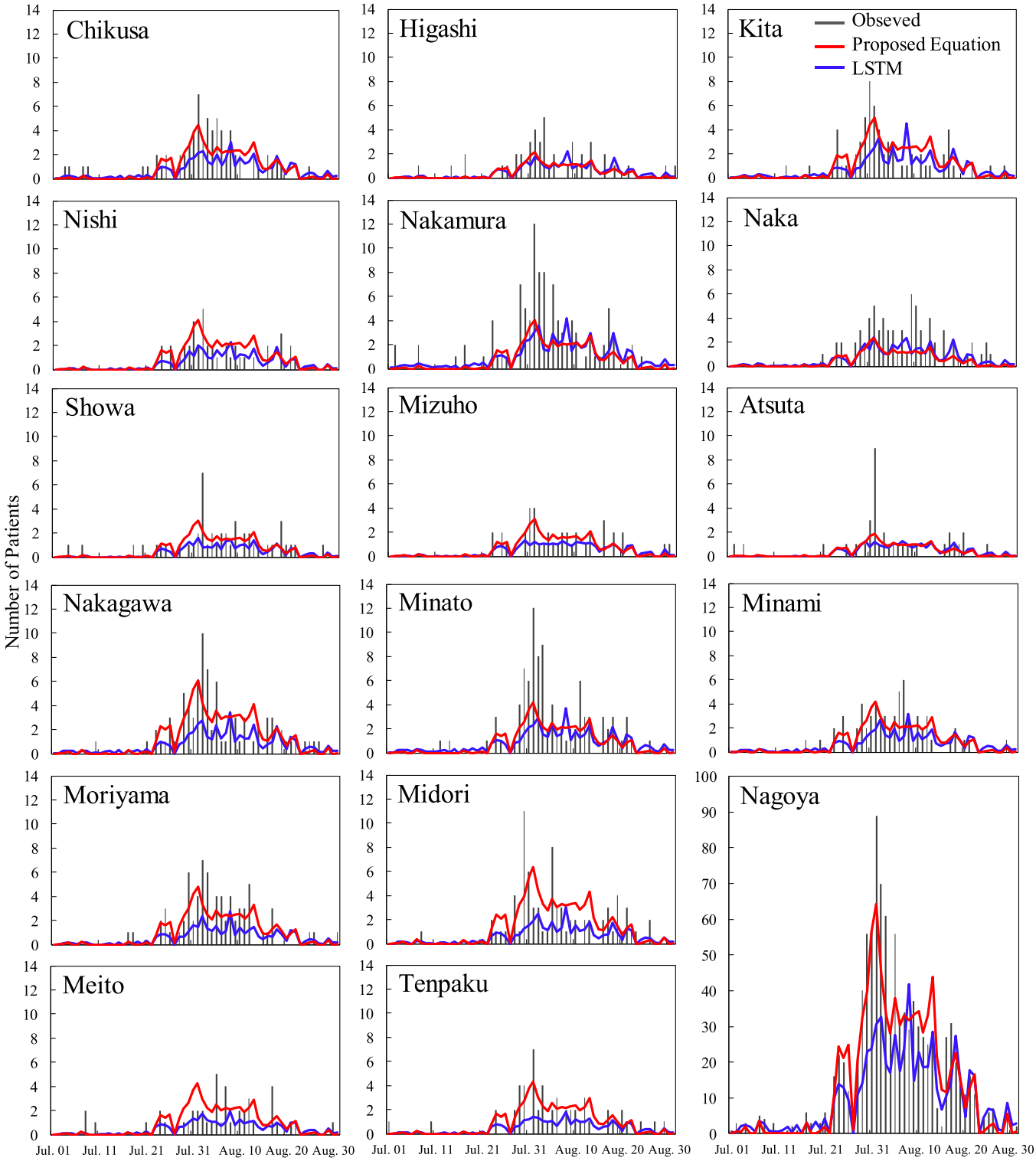

Fig. 6. Estimated number of patients in each ward of Nagoya City using equations and LSTM compared with actual values (year: 2019).

combining the prediction techniques of climate change and age-specific population distribution. These findings would lead to public awareness and decision-making support for long-term government policies.

There are two primary limitations of this study. First, the regression analysis and LSTM were conducted using the dataset in one city as a case study because the only available dataset currently, including the detailed locations, is the data from Nagoya City. However, the estimation in a similar approach over other prefectures such as Tokyo was shown to be highly accurate (Kodera et al., 2019), suggesting that the proposed frameworks are applicable to other regions with minor parameter tuning. Second, we assumed that the number of patients with a heat-related illness was the same as the number of individuals transported by an ambulance owing to a heat-related illness. In Japan, ambulance transportation can be used for free, and the number of transported people and patients are comparable (Shimazaki et al., 2020).

Table 5
Daily residual error averaged over six years in each ward.

Ward name	Proposed equation (number of people)	LSTM (number of people)
Chikusa	0.82	0.80
Higashi	0.55	0.50
Kita	0.80	0.71
Nishi	0.68	0.59
Nakamura	1.03	0.9
Naka	0.81	0.71
Showa	0.59	0.53
Mizuho	0.55	0.53
Atsuta	0.45	0.39
Nakagawa	0.86	0.79
Minato	0.89	0.89
Minami	0.72	0.72
Moriyama	0.73	0.70
Midori	0.86	0.73
Meito	0.71	0.54
Tenpaku	0.71	0.64
Average	0.74	0.67

Table A1
Daily residual error in estimating number of patients transported from indoor and outdoor locations in each ward by proposed equation over six years (2014–2019).

	Indoor (number of people)	Outdoor (number of people)
Chikusa	0.51	0.51
Higashi	0.31	0.38
Kita	0.53	0.55
Nishi	0.42	0.49
Nakamura	0.57	0.73
Naka	0.31	0.67
Showa	0.40	0.37
Mizuho	0.35	0.41
Atsuta	0.25	0.32
Nakagawa	0.62	0.58
Minato	0.50	0.67
Minami	0.46	0.48
Moriyama	0.52	0.50
Midori	0.53	0.60
Meito	0.41	0.49
Tenpaku	0.42	0.47
Average	0.44	0.51

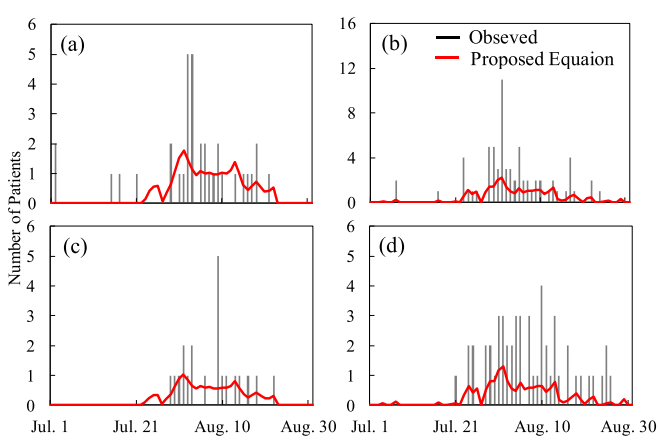

Fig. A1. Example of the estimated number of patients by the proposed equation compared with true values, (a) Nakamura ward (indoor), (b) Nakamura ward (outdoor), (c) Naka ward (indoor) and (d) Naka ward (outdoor). Data indicates summer 2019.

The approximation made by these frameworks would be an alert indicator of for heat stroke that considers differences in certain factors such as climate, heat adaption, and the EMS, including in areas where data accumulation is difficult, such as in developing countries.

5. Conclusion

Two frameworks were proposed for estimating the number of patients with heat-related illness: regression equation and LSTM. Our analysis suggests that patients with heat-related illnesses transported from indoor and outdoor locations are characterized by daily average temperatures over the previous 50 and 20 days, respectively, for heat adaptation. The proposed equation obtained a highly accurate estimate of daily heat-related illness that was comparable to that of the LSTM model. These frameworks can make estimations using only the ambient temperature and population distribution by age available online.

Author Contributions

AH and ER conceived the idea. MN and AH presented the idea of the appendix. TN, SK and ER developed the analytical formulation and performed the computations. SK and ER verified the analytical methods. HS, RK, and KW acquired and analyzed the data of ambulance dispatches. All authors have discussed the results and contributed to the final manuscript.

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Declaration of Competing Interest

The authors declare no competing interests.

Table A2
Daily residual error in Nakamura and Naka wards using proposed equation (Year: 2019).

	Nakamura	Naka
Indoor	0.49	0.28
Outdoor	0.77	0.65

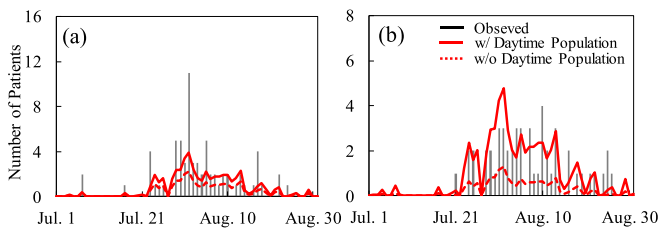

Fig. A2. Number of estimated patients transported from outdoor locations compared to its true value in (a) Nakamura and (b) Naka wards (Year: 2019). Solid and dotted lines indicate the results with and without considering a daytime population coefficient.

Table A3
Daily residual error with the coefficient of daytime population in Nakamura and Naka wards (Year: 2019).

	Without coefficient	With coefficient
Nakamura	0.77	0.65
Naka	0.65	0.51

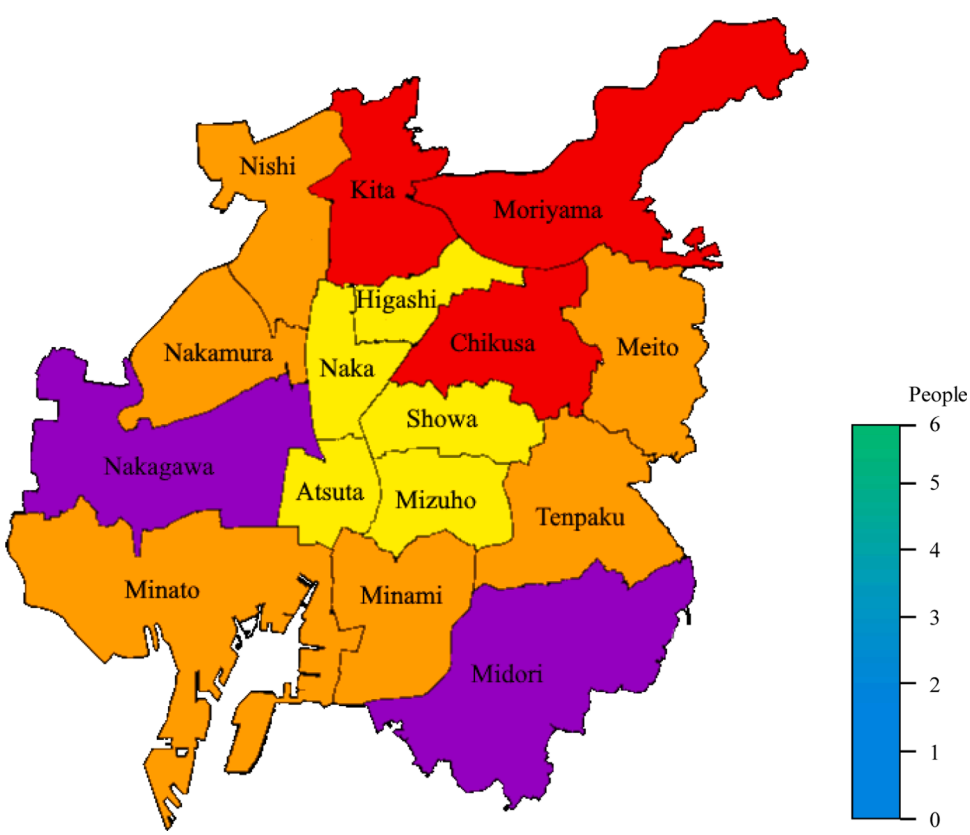

Fig. B1. Colored map of expected number of heat-related-illness patients on August 18, the day with the highest number of patients (65 people) in 2020.

Appendix A. Estimation of Nakamura and Naka Wards

We proposed an equation for estimating the number of patients with an accuracy comparable to that of LSTM. The difference in the mean residual errors in the entire city for these two methods was comparable; however, the mean daily residual errors were larger for Nakamura, Minato, and Midori wards than those for the remaining wards. The largest errors were observed in the outdoor estimation of the Nakamura and Naka wards (see Table A1). Fig. A1 shows the comparison between the estimations and actual data of number of patients transported indoors and outdoors, and Table A2 lists the mean daily residual errors. The outdoor estimation error was larger than the indoor estimation error.

Nakamura ward is located in the center of the city, and a similar trend was observed in the adjacent Naka ward. Urban wards are characterized by a high proportion of heat-related-illness patients transported from outdoor locations. The ratio of patients transported from outdoor locations was 60.1% and 76.5% between 2014 and 2019 in Nakamura and Naka wards, respectively, compared to 55% in the entire city of Nagoya. The daytime number of patients in Nakamura and Naka wards increased by 172% and 364%, respectively, compared to 113% in Nagoya City (Nagoya City, 2015). In the wards where the daytime population decreases, such as Nakagawa (88%) and Midori (79%), the estimates tend to exceed the actual data owing to population outflow (see Fig. 6). The number of patients transported from outdoor locations was 206% higher than that estimated by the equation in the Nakamura ward and 306% higher in the Naka ward. These values were close to the daytime population inflow. Based on these findings, we introduced a coefficient q (=1.72 in Nakamura ward and 3.64 in Naka ward) considering daytime population inflow in the estimating equation for patients transported from outdoors as follows:

$$y_{outdoor}(x) = q \cdot a_{out} \cdot \left[e^{k_{out} \cdot x_0} + l_{out} \right] \cdot \sum_n \{P(n) \cdot (be^{cn} + d)\}. \tag{6}$$

The mean daily residual error decreased in both the Nakamura and Naka wards owing to the introduction of the q coefficient (Fig. A2 and Table A3). The accuracy of the estimation equation proposed here will increase if combined with the daily population inflows and outflows in a future study.

Appendix B. Application of the data for public awareness

The data provided here was used for public enlightenment and preliminary implementation in the arrangement of ambulances. The colored map of the expected number of patients with a heat-related illness is shown in Fig. B1 was presented in a television program (Nagoya Broadcasting Network Co., Ltd.) for two weeks (August 10–23) during the hottest period of the year. Although it is difficult to evaluate its impact due to the COVID-19 pandemic in the summer of 2020, the actual number of patients was 7.3% lower than the estimated value. Notably, the actual value in the remaining period was comparable to the estimated value.

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